

§III Ermentrout-Kopell model (1986)

- also called θ -model or theta-model
- one dimensional, but mathematically nicer than LIF
- *very* abstract
- mathematical simplicity $\rightarrow$ useful to develop intuition about large networks

§III(a) motivation

Recall why 1d models don't make AP:

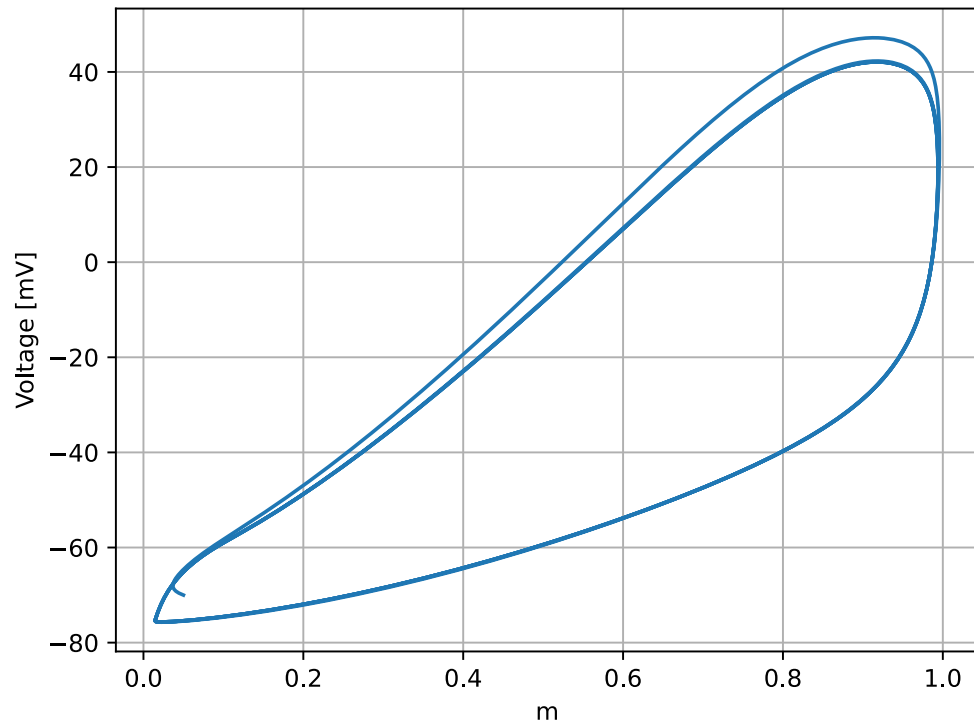
- Want this $V(t)$ as a solution:



- look at the trajectory of an AP in 1D phase space
- so 1st order systems can not make an AP spike *unless we include threshold / reset by hand*
- note:

Recall how AP looks in HH, projected to two-dimensions

- tonic spiking, $I = 10$ phase space pic



- allows $dV/dt > 0$ and $dV/dt < 0$ at same V , because of different values of gate variable m
- phase space is 4D, but don't 'feel' all of it - spend most time on a loop

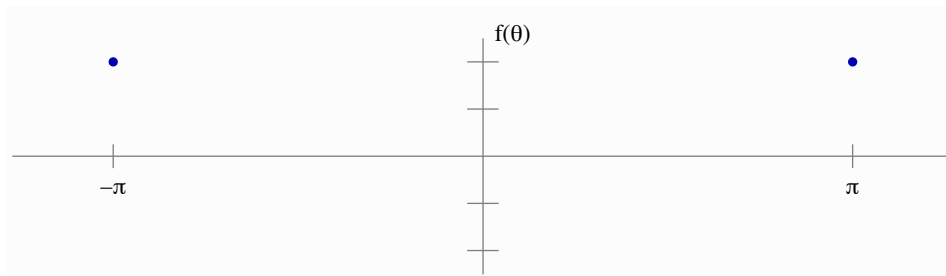
§III(b) the θ -neuron model

$$\frac{d\theta}{dt} = f(\theta) \quad \text{where}$$

- $f(\theta) = (1 - \cos \theta) + (1 + \cos \theta) a$
- $-\pi \leq \theta < \pi$
- $\theta = \pm\pi$ counts as ‘firing’

rough description:

What does $f(\theta)$ look like?

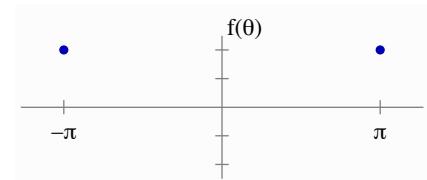


§III(c) Analysis and dynamics

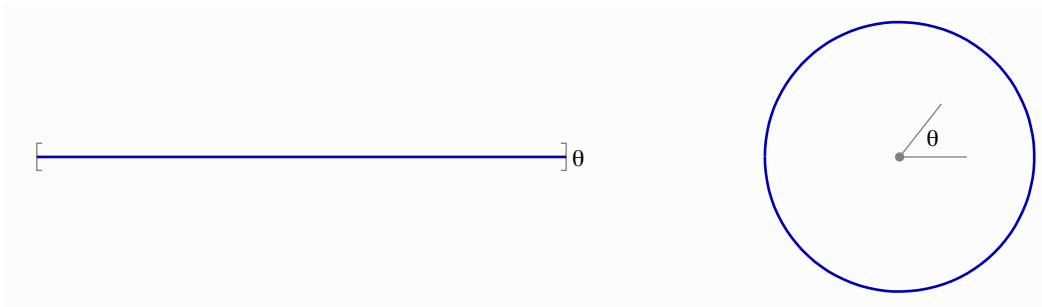
- phase space =
- expect different dynamics in $a < 0$ and $a > 0$ (*why?*)

case $a < 0$

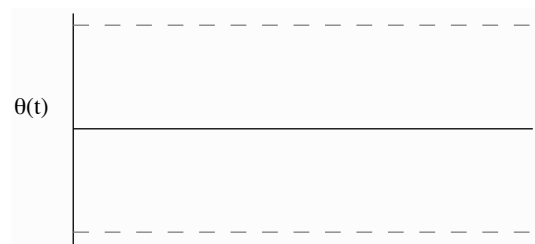
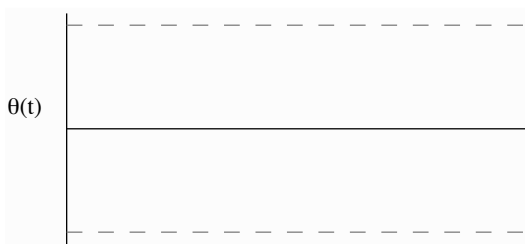
- $f(\theta)$ has two zeros, at $\theta_{\text{fp}} = \pm \cos^{-1} \left(\frac{1+a}{1-a} \right)$



- phase space sketch:

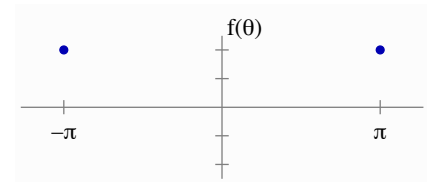


- f.p. in $\theta < 0$ is stable
- f.p. in $\theta > 0$ is unstable
- time-series:

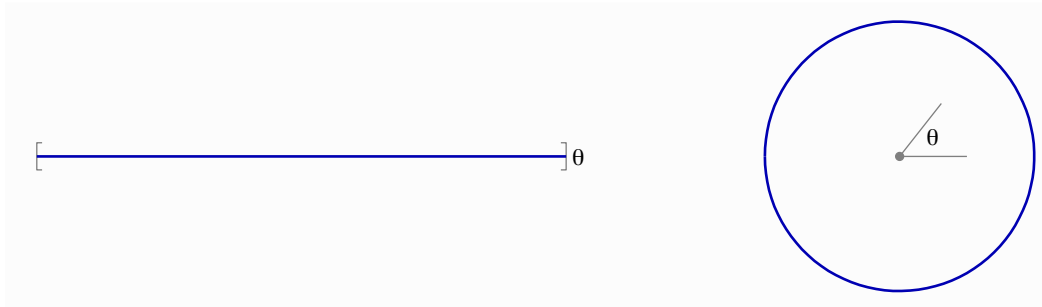


case $a > 0$

- $f(\theta)$ has no fixed points

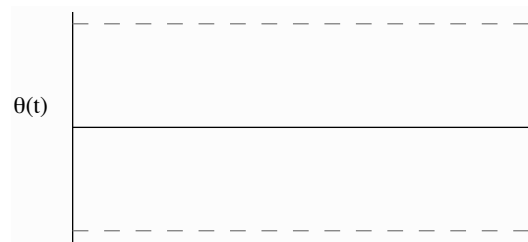


- phase space sketch:



- no fixed points

- time-series:



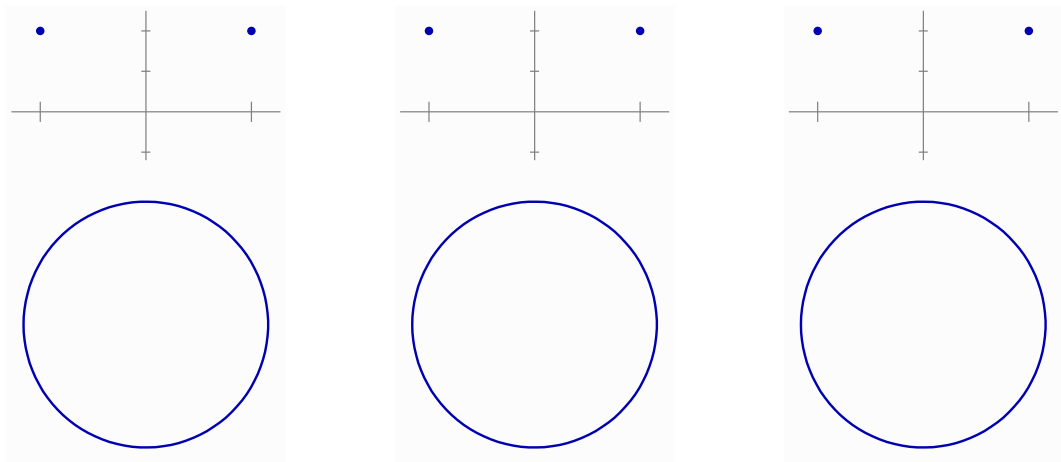
The θ -model is mathematically simple, and the behavior of a θ -neuron intuitive. It can be used as to build intuition of networks of interacting neurons. We will use it as an example to explore the transition from *excitable* to *continuous firing*.

§III(d) Example of a bifurcation in a dynamical system

bifurcation = a sudden change in the qualitative model behavior (e.g., the number of fixed points) in response to change in a parameter

Consider the θ -neuron as parameter a increases:

$$\frac{d\theta}{dt} = f(\theta) = (1 - \cos \theta) + (1 + \cos \theta) a$$



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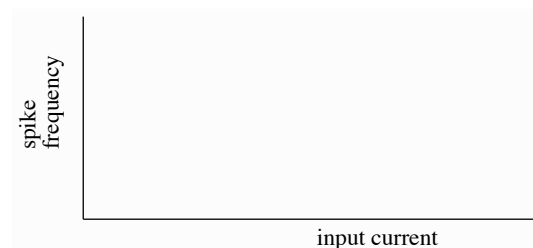
§IV Bifurcations in dynamical systems

bifurcation = a sudden change in the qualitative model behavior (e.g., the number of fixed points) in response to change in a parameter

Example: excitable neuron continuously firing neuron

FI curve:

One classification of neurons is based on the frequency response of continuously firing neurons



- includes low frequencies
- large frequency range
- typical of regular spiking (RS) neurons
- Ex:



- no low frequencies
- smaller bandwidth
- typical of fast spiking (FS) neurons
- Ex:

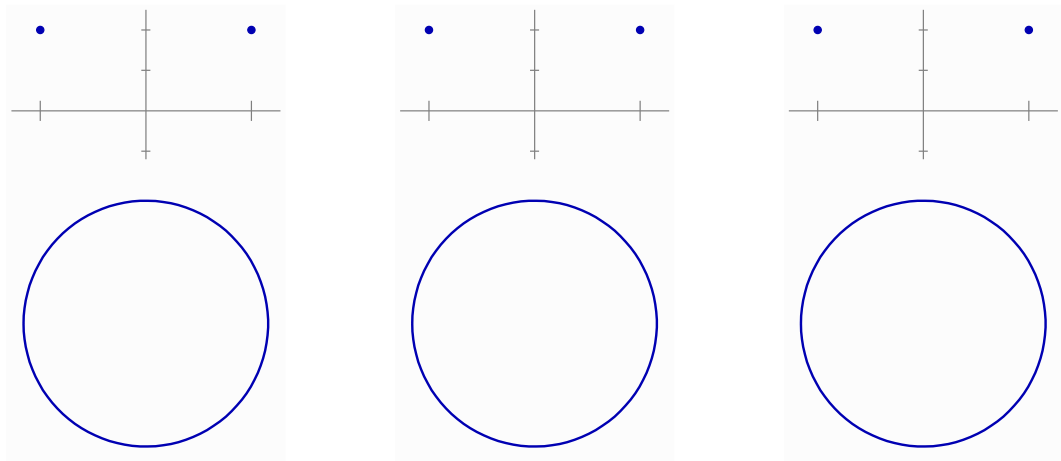
Many types of bifurcations exist - we consider one example.

§IV(a) Bifurcation diagram

Example:

For the θ -neuron:

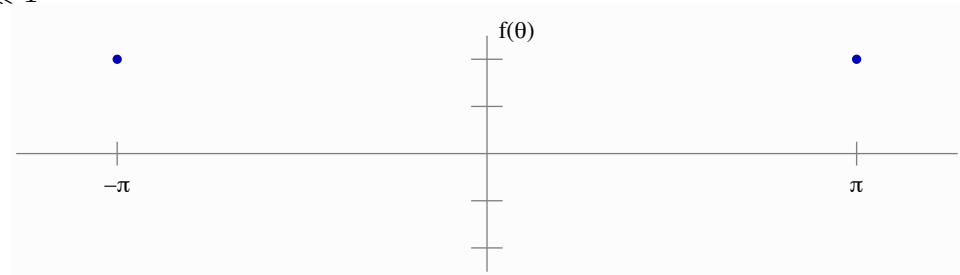
$$\frac{d\theta}{dt} = f(\theta) = (1 - \cos \theta) + (1 + \cos \theta) a$$



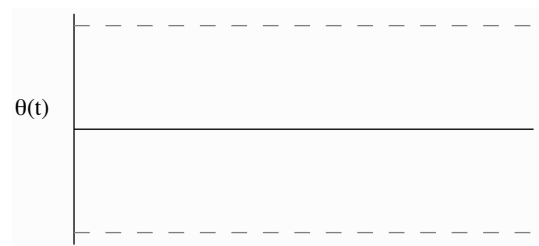
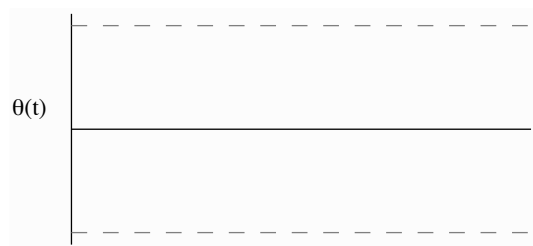
bifurcation diagram:

Is this Class I or Class II behavior?

consider $0 < a \ll 1$



so neuron fires continuously



Conclusion:

Closing remarks

- Modern computers are fast: in practice, we can use HH-style neurons.
- HH-style models can generate Class I or Class II frequency response, and *lots* of other exotic behavior, either as a single compartment or when combined in a network.
- Other models (θ) are useful mathematical tools to develop intuition about neurons, and insight into their behavior.